ACM Multimedia 2003 Tutorial Proposal

Title: Machine Learning for Multimedia Applications (Full Day)

Tutor: Edward Chang, University of California, Santa Barbara

Target Audience

Broad audience including students and researchers interested in an overview of a suite of new generation Machine Learning algorithms for analyzing high-dimensional imagery and video data. No particular background expected.

Goals and Motivation

Bridging low-level perceptual features and high-level semantics is a pivotal research problem that exhibits many challenges. In order to develop useful and scalable solutions for bridging the feature-and-semantics gap, researchers should comprehend the fundamental problems and become familiar with recently developed Machine Learning techniques. This tutorial will first point out the pitfalls and fallacies of applying some traditional statistical methods and information retrieval techniques to multimedia applications. We will then introduce a suite of new Machine Learning algorithms for addressing the feature-to-semantics mapping problem in a principled way.

The principal design goal of a visual information retrieval system is to return data (images or video clips) that accurately match users' query concepts. To achieve this design goal, the system must first comprehend a user's query concept thoroughly, and then find data that accurately match the concept in the low-level input space. Statistical learning techniques can assist achieving the design goal via two complementary avenues: *semantic annotation* and *query-concept learning*.

Semantic annotation provides visual data with semantic labels to support keyword-based searches. Several studies propose semi-automatic annotation methods to propagate keywords from a small set of annotated images to the other images. Although semantic annotation can provide some relevant query results, annotation is often subjective and narrowly construed. When it is, query performance may be compromised. To thoroughly understand a query concept, with all of its semantics and subjectivity, a system must learn the target concept from the user directly via *query-concept learning*. (*Semantic annotation* can assist, but not replace, *query-concept learning*.)

At first glace, it would seem that traditional supervised learning methods such as neural networks, decision trees, and Support Vector Machines (SVMs) could be used directly to perform *semantic annotation* and *query-concept learning*. Unfortunately, "classical" learning algorithms are not adequate to deal with the technical challenges posed by these two tasks. To illustrate, let D denote the number of low-level features, N the number of training instances, N^+ the number of positive training instances, and N^- the number of negative training instances ($N = N^+ + N^-$). Two major technical challenges arise:

1. *Scarcity of training data.* The features-to-semantics mapping problem often comes up against the $D > N$ challenge. For instance, in the query-concept learning scenario, the number of low-level features that characterize an image (D) is greater than the number of images a user would be willing to label (N) during a relevance feedback session. As pointed out by D. Donoho, the theories underlying "classical" data analysis are based on the assumption that $D < N$ and $N \to \infty$. But when $D > N$, the basic methodology which was used in the classical situation is not strictly applicable.

2. *Imbalance of training classes.* The target class in the training pool is typically outnumbered by the non-target classes ($N^- >> N^+$). For instance, in a k-class classification problem with each class having the same number of training instances, the target class is outnumbered by the non-target classes by a ratio of $k - 1 : 1$. The class boundary of imbalanced training classes tends to skew toward the target class when k is large. This skew makes class prediction less reliable.

This tutorial will present emerging machine learning methods for dealing with the above challenges. This full-day tutorial will be organized into five parts.

Outline

1. *Classical Methods.* (1.0 hour)
   - Fundamental problems.
   - Machine decision theories.
   - The least squares, the nearest neighbors, and anything in between.
2. *Emerging Methods.* (2.0 hour)
   - Generative methods.
     - The Hidden Markov Model (HMM).
     - The MEGA algorithm.
     - Active learning.
   - Discriminative methods.
     - Support Vector Machines.
     - Bayes Point Machines.
     - Gaussian process.
     - General Kernel methods.
3. *High Dimensionality.* (1.0 hour)
   - Curses of dimensionality.
   - Dimension reduction techniques.
   - Indexing techniques.

- Pitfalls.
  - Is it more important to acquire positive-labeled instances than to acquire negative-labeled ones?
  - Can transductive learning work when D is large and $N^- >> N^+$?

4. *Advanced Topics.* (1.0 hour)
   - Imbalance dataset learning.
   - Sequence data learning.
   - Kernel design and composition.
   - Kernel target alignment.
5. *Demos and Round-table Discussion.* (1.0 hour)

Tutor

Prof. Edward Chang received his Ph.D. in Electrical Engineering at Stanford University in 1999. He joined the University of California at Santa Barbara as an Assistant Professor in 1999, and he has been a tenured Associate Professor of Electrical & Computer Engineering and of Computer Science since 2003. His current research interests include Statistical Learning and Multimedia Databases. His work related to this tutorial subject has been extensively published in the proceedings of some premier conferences including ACM Multimedia (Multimedia), ICML (Machine Learning), ICDM (Data Mining), ICIP (Image Processing), and CVPR (Computer Vision and Pattern Recognition). (Please refer to http://www-db.stanford.edu/ echang/chronological-order.html for a complete list of his publications.) Prof. Chang's work on Perception-Based Image Retrieval (PBIR), which applies statistical learning to formulate users' subjective similarity metrics, was recognized as a major breakthrough in image retrieval by the CBIR panel at the 2002 IEEE International Conference on Multimedia. He is a recipient of the IBM Faculty Partnership Award from 2000 to 2002, and the NSF Career Award in 2002. Prof. Chang is a co-founder and the CTO of VIMA Technologies.

Prior Tutorial Experience

The first half of this tutorial was given by Prof. Chang at the 8th International Conference on Database Systems for Advanced Applications (DASFAA), held in Kyoto University, in March 2003. The tutorial drew more than 150 attendees, and was commended by professor Jeff Ullman (Stanford) as very effective. The same tutorial will be given at ICME at Baltomire this July. The quality of this proposed tutorial is expected to be uncompromised at ACM Multimedia.

AV Equipment Requirements

Laptop projector; microphone.